AUDIO2TAB: END-TO-END AUTOMATIC GUITAR TABLATURE TRANSCRIPTION

Jackson Loth[*] *Pedro Sarmento*[†] *Mathieu Barthet*[‡*]

[*] Centre for Digital Music, Queen Mary University of London
[†]Moises.AI
[‡]Aix-Marseille Univ CNRS PRISM

ABSTRACT

We present Audio2Tab, a model and methodology for the task of automatic guitar tablature transcription (AGTT). We approach the task as a sequence modelling problem, predicting a sequence of tokens representing a guitar tablature given a 15-second audio recording of a guitar performance. Our model can predict both string/fret and expressive playing techniques. The predicted tokens can also be easily decoded into a tablature score and viewed using common guitar tablature software such as GuitarPro. We fine-tuned an open-source speech-to-text model in two stages, first on a large amount of synthetic data and then a smaller amount of high-quality real guitar recordings. We demonstrate the effectiveness of our methodology through both sequence and frame-level AGTT experiments.

Index Terms— automatic guitar tablature transcription, automatic music transcription, sequence-to-sequence modelling, guitar tablatures

1. INTRODUCTION

The guitar is a popular instrument in modern Western music and can be heard across multiple contemporary music recordings. The instrument features six strings which can produce different pitches when plucked depending on the location (i.e. the fret) they are held at. Thus, the same pitch can be played in multiple different locations. Tablatures, often called tabs, have become a common way to represent music on stringed instruments such as guitars as they excel at intuitively explaining how a piece of music is played (see Figure 1). It does this through representing notes as combinations of strings and frets, and can also represent various expressive techniques such as bends and hammer-ons [1].

Due to the popularity of tabs for learning music, transcribing a guitar performance to tab is an extremely valuable task. Automatic guitar transcription (AGT) is a growing subfield of automatic music transcription (AMT) which aims to predict the notes played in a guitar performance, with recent research achieving impressive results [2] [3] [4] [5]. Further predicting the string/fret played is referred to as automatic guitar tablature transcription (AGTT) and has seen a growing amount of interest. Most previous works predicts note onsets or activations for specific string and fret combinations [6] [7] [8] [9], with more recent works extending to basic expressive elements [10]. However, these approaches are still limited in their ability to create a readable tablature score. While plenty of score transcription works exist [11] [12] [13] [14], they are typically framed more as a sequence-to-sequence task and rarely focus on guitar, much less guitar tabs.

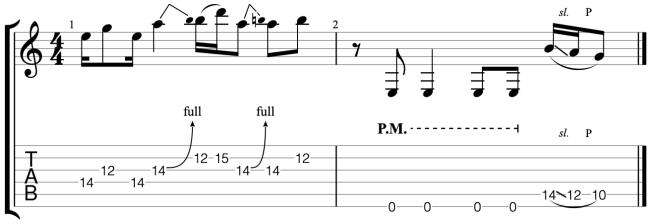


Fig. 1. A guitar phrase in traditional score (top; represents an example of the output of an AMT/AGT system) and tablature (bottom; an example of the output of an AGTT system).

Our approach, which we name Audio2Tab, differs from most AGTT works by predicting structured tablature scores, including tempo and expressive techniques, using a single model based on Whisper [15], a pretrained speech-to-text model. Inspired by score transcription works, Audio2Tab predicts token sequences from audio using a tokenization based on DadaGP [16] and automatically converts the sequences to the common `.gp5` format for use in guitar tablature software. In this paper, we present: (1) a formulation of AGTT as a sequence-to-sequence task and a methodology for training Audio2Tab using a combination of synthetic and real data; (2) a full evaluation of the model performance, including note, string/fret, expressive technique and tempo prediction covering several ablations; (3) a comparison to other AGTT models in a frame-level prediction task; (4) a public repository[1], audio examples[2], and checkpoint for Audio2Tab which can be used for inference, benchmarking, or further training.

JL is supported by the EPSRC UKRI Centre for Doctoral Training in Artificial Intelligence and Music (Grant no. EP/S022694/1).

[1]GitHub repository
[2]Audio and tablature examples

We hope that this work provides a useful benchmark for researchers and lead to useful tools for guitarists when learning new songs or transcribing their own music.

2. METHODOLOGY

2.1. Data

We make use of both synthetic data and real guitar data in our work. For synthetic data, we use SynthTab, a large dataset of synthetic guitar audio and tablature annotations [17]. This dataset allows the model to learn an initial prior before fine-tuning it on more realistic data. For realistic guitar data we use the GOAT [18] dataset, a recent guitar dataset with realistic guitar audio and aligned tablature annotations. Furthermore, we use pitch and tempo-based data augmentations while training our models. For pitch augmentation, we apply a random pitch shift between -3 and 3 semitones to each batch and update the tablature annotation accordingly. If this pitch shift produces invalid fret coordinates for any sample in an audio chunk (such as frets below 0 or above 24), the augmentation is skipped entirely for those affected instances. For tempo augmentation, we add between 0 and up to $\pm$ 10 bpm at random to the tempo of each chunk in the batch. We then use the difference between the original and new tempo to calculate the exact percentage of time stretching to be applied to the audio. In total, we use 1035.6 hours of synthetic data (with 258.9 hours corresponding to unique tabs) and 4.5 hours of real data.

2.2. Tablature Annotation

The tablature annotations in both datasets are converted to DadaGP-like sequences. We adopt a simplified version of the DadaGP tokenization which omits some features such as artist and timbre information. This allows us to define a specific vocabulary which can then be learned by the model. Predicting DadaGP sequences also has the benefit of being able to be easily converted into a `.gp5` file with the DadaGP decoder, which can then be opened in guitar tablature software such as GuitarPro. Finally, the DadaGP encoding also represents many expressive playing techniques such as hammer-ons and bends. These are typically represented using `nfx` tokens, with some beat-level effects represented by `bfx` tokens.

2.3. Model

We use Whisper [15], a pretrained transformer-based encoder-decoder speech-to-text model, as the basis for our tablature transcription model. Specifically, we use the `whisper-small` model checkpoint. This was motivated by our formulation of AGTT as an audio-to-sequence task, with Whisper providing a natural architectural fit as well as a strong pretrained initialization. Unless otherwise stated, all of our models are fine-tuned from this checkpoint. We use this architecture as is, using the same input representation (log mel-spectrograms) and the same autoregressive decoding strategy. Because our vocabulary consists of only DadaGP tokens rather than natural language tokens, we define a custom tokenizer for the task. This tokenizer maps each token in our DadaGP vocabulary to a unique integer, which can then be used as the target for the model to predict.

2.4. Training

We split both datasets into 90% training, 5% test, and 5% validation splits. Each split was divided by song to prevent song overlap between splits. For the smaller GOAT dataset, we designed the splits to maintain similar expressive-technique token distributions[3]. SynthTab was split randomly by song, relying on its larger size to support balanced distributions. Training comprised two stages. First, we pretrained on SynthTab to learn the DadaGP structure and map pitches and expressive techniques to their corresponding tokens, removing `nfx` tokens for techniques absent from the audio rendering. For each sampled song, we randomly selected a 15-second audio segment and its corresponding DadaGP sequence. Segments began at note onsets to avoid partial-note starts and complex timing-offset representations while preserving rhythmic quantisation. Second, we fine-tuned the SynthTab checkpoint on GOAT. To maximise coverage of this smaller dataset, we extracted 15-second segments using a one-second hop, shifting each window's start to the next note onset. We also trained a GOAT-only model to assess the contribution of synthetic pretraining. All models used a batch size of 32. SynthTab pretraining used a learning rate of 10^{-5}, while GOAT fine-tuning and GOAT-only training used 10^{-6}, as the higher rate caused instability and poorer performance. Checkpoints were selected by the lowest validation loss.

3. RESULTS AND DISCUSSION

3.1. Sequence-Level Transcription

Audio2Tab was trained on 15-second audio windows, thus our first experiment evaluates performance under the same constraint. Unfortunately, to our knowledge there are no comparable models which frame AGTT in this way at the time of writing.

We present an ablation experiment which both validates our choice to use the Whisper speech checkpoint as well as demonstrates the general performance of the model. We also adopt an evaluation framework similar to MV2H [19], specifically the non-aligned variant [20], by using a recent score-performance alignment method [21] to align the predicted notes to the ground truth notes based on note pitch and onset. This allows us to address the impact of sequence insertions/deletions disrupting typical time-based transcription metrics. From this alignment, we calculate F1 scores for pitch onset (matching notes with the same pitch), string/fret onset (matching notes with the same string/fret value), FX

[3] This is different from the default train/test splits in the published dataset.

Model	Test Set	Pitch F1 ↑	String/Fret F1 ↑	FX F1 ↑	Note Value ↑	WER ↓	Tempo ACC_1 ↑	Tempo ACC_2 ↑
Audio2Tab-S	SynthTab	**80.36**	**64.21**	**20.86**	**90.59**	**43.86**	**77.55**	**84.76**
Audio2Tab-G	SynthTab	10.19	6.44	3.69	33.70	105.52	1.60	23.47
Audio2Tab-SG	SynthTab	76.81	58.31	16.90	88.02	52.64	60.76	75.98
Audio2Tab-RSG	SynthTab	54.18	35.49	10.15	76.28	79.43	14.21	20.83
Audio2Tab-S	GOAT	58.23	33.17	3.91	69.26	117.61	36.36	70.91
Audio2Tab-G	GOAT	12.84	10.60	8.99	29.91	118.64	0.00	0.00
Audio2Tab-SG	GOAT	**80.82**	**61.91**	**31.65**	**88.05**	**55.91**	**67.27**	**90.91**
Audio2Tab-RSG	GOAT	58.06	37.24	7.24	79.00	81.01	0.00	0.00

Table 1. Sequence-level tablature transcription results on the SynthTab and GOAT test sets. Pitch F1 measures note pitch accuracy, String/Fret F1 evaluates string and fret accuracy, and FX F1 evaluates performance technique recognition. Note value is calculated only on correctly matched pairs in pitch. All F1 metrics are calculated using aligned note pairs. All metrics are scaled by 100 for consistency. Best results are shown in bold.

(matching notes with the correct expressive technique tokens) and note value (how closely note duration matches in aligned notes, calculated the same way as MV2H). Unaligned notes become false negatives/positives. In addition to these metrics, we also calculate word error rate (WER) and two tempo accuracy measurements commonly used in tempo prediction literature [22]. ACC_1 checks whether the predicted tempo is within $\pm 4\%$ of the ground truth tempo, while ACC_2 allows errors to be off by factors of 2, 3, $1/2$ and $1/3$.

We report results for the models trained only on SynthTab (Audio2Tab-S), trained only on GOAT (Audio2Tab-G) and trained on both SynthTab and GOAT (Audio2Tab-SG). We also train a variant of Audio2Tab-SG in which the model is initialised using random weights before the SynthTab training stage, rather than using the pretrained weights from Whisper. This allows us to test the effect of the pretrained weights from Whisper on the performance of the model. We refer to this model as Audio2Tab-RSG. Each model is evaluated on the test splits of SynthTab and GOAT (see Table 1). We also present a breakdown of the F1 score for several categories of expressive techniques using the SynthTab + GOAT model evaluated on the test split of GOAT (see Table 2). A small fraction of SynthTab examples could not be aligned and were excluded from evaluation ($< 1\%$ for Audio2Tab-S and Audio2Tab-SG; $\approx 1.7\%$ for Audio2Tab-G). No GOAT examples were excluded. All audio segments begin at note onsets for consistency with training.

Technique	F1 ↑
Bends	14.63
Hammer-ons/pull-offs	64.37
Slides	67.69
Palm-mutes	67.19
Dead notes	20.74

Table 2. Expressive technique prediction using Audio2Tab-SG evaluated with F1 scores on the test split of the GOAT dataset, calculated using the aligned note pairs.

Audio2Tab-SG achieves the strongest overall performance on real guitar recordings, with a pitch F1 of 80.82%, string/fret F1 of 61.91%, and note-value score of 88.05. The lower string/fret performance reflects the additional ambiguity of predicting fretboard position rather than pitch alone. Replacing predicted string/fret tokens with the most common position for each pitch reduced F1 to 44.39% on GOAT and 35.27% on SynthTab, suggesting that the model learns contextual fingering information rather than relying only on dataset priors. Expressive technique prediction is more uneven: hammer-ons/pull-offs, slides, and palm-mutes are predicted relatively well, while bends and dead notes remain difficult. Bends are likely often confused with sequences of separate notes, while dead notes are absent from SynthTab and therefore must be learned from the comparatively small GOAT dataset. Although aligned note metrics are strong, the WER of 55.91 indicates that insertion, deletion, and substitution errors remain common.

The ablations demonstrate the importance of both fine-tuning stages. Training only on GOAT performs poorly across all metrics, while training only on SynthTab transfers imperfectly to real recordings. In contrast, synthetic training followed by GOAT fine-tuning substantially improves real-data performance while largely retaining performance on synthetic data, indicating that synthetic data provides a useful prior but real recordings are necessary for domain adaptation. Initialising from Whisper is also beneficial: Audio2Tab-SG outperforms the randomly initialised Audio2Tab-RSG model on every metric on both datasets, showing that speech-derived pretrained representations transfer effectively to guitar tablature transcription. Finally, Audio2Tab-SG achieves strong relaxed tempo accuracy (ACC_2) on GOAT, although because tempo is predicted implicitly as part of the token sequence, it is unclear how much of this performance reflects learned tempo estimation versus dataset priors.

3.2. Frame-Level Transcription

As our sequence-level transcription does not have comparable models, we test how our model performs in a more typical

Model	PP ↑	PR ↑	PF ↑	TP ↑	TR ↑	TF ↑	TDR ↑	FNR ↓	FPR ↓
TabCNN	75.3	59.4	66.4	74.4	58.7	65.6	98.7	**29.3**	4.7
FretNet	68.9	65.0	66.9	68.1	64.3	66.2	98.9	17.1	6.6
Noise2Fret	**82.8**	**77.4**	**80.0**	**82.2**	**76.9**	**79.5**	**99.3**	16.5	**2.2**
Audio2Tab	79.6	59.8	68.3	44.5	33.4	38.2	55.9	45.7	4.7

Table 3. Frame-level tablature transcription results. All metrics are scaled by 100 for consistency. Best results within each test set are shown in bold. The metrics presented are pitch precision (PP), pitch recall (PR), pitch F1 (PF), tab precision (TP), tab recall (TR), tab F1 (TF), tab disambiguation rate (TDR), false negative rate (FNR) and false positive rate (FPR). Results for TabCNN, FretNet, and Noise2Fret are taken from [10]; Audio2Tab results are from our implementation.

frame-level transcription context in order to compare against prior AGTT models. We limit the audio window to 100 ms and evaluate on the same task as Noise2Fret [10] by targeting up to two sequential note events in the audio frame and calculating several tablature metrics. All audio windows start at note onsets. For this experiment, we use the GOAT train/test splits employed by Noise2Fret rather than the sequence-level split described in Section 2.4. This task considers note pitch and string/fret values, excluding timing, duration, tempo and expressive techniques. We compare against baselines calculated for TabCNN [6], FretNet [7] and Noise2Fret using the results presented in [10]. Results are presented in Table 3. While our model does not achieve the high performance of Noise2Fret, it remains competitive in the pitch-based metrics compared to TabCNN and FretNet. It struggles significantly more in the tab-based metrics however, which we hypothesise as being due to having significantly less context compared to the sequence-level task. We suspect that the short-context frame-level task may favour specialized architectures such as Noise2Fret, whereas Audio2Tab is designed primarily for the broader sequence-level task.

4. LIMITATIONS & FUTURE WORK

Despite promising results, our evaluation has some limitations. Our sequence-level evaluation is based on an extra alignment step, which could introduce some bias into the reported results. Subjective evaluation by guitarists would further clarify transcription quality. Our model was also only trained on DI electric guitar, therefore it is unknown how well it generalises to other guitar timbres such as acoustic or distorted electric guitar. GOAT's predominantly rock-based recordings and our focus on rock-oriented expressive techniques may also limit generalisation to genres such as jazz and classical guitar. Finally, our model transcribes segments of up to 15 seconds, while Whisper accepts inputs up to 30 seconds; neither accommodates typical song lengths. Longer recordings could be transcribed by concatenating chunk-level outputs, but maintaining consistent tempo and smooth transitions between chunks would require further investigation.

5. CONCLUSION

In this paper we propose Audio2Tab, a baseline for the task of end-to-end automatic guitar tablature transcription. We repurposed and trained a Whisper model, leveraging the text-like representation of tablatures present in the GOAT dataset. We studied training strategies such as a preliminary synthetic fine-tuning step, relevant data augmentation techniques, and fine-tuning approaches on an real-world dataset. The results demonstrate the validity of this approach, with our best model obtaining F1 scores of 80.82 for pitch prediction and 61.91 for string and fret prediction, on a model first fine-tuned on synthetic data (SynthTab dataset) and further fine-tuned on real-world data (GOAT dataset). We also demonstrate the validity of this pretraining and fine-tuning approach, showing how it improves performance on real guitar audio while maintaining performance on synthetic audio. Our best model can predict common expressive techniques such as hammer-ons, pull-offs, slides and palm-mutes relatively well, as well as accurately predict tempo. We hope that this work proves useful for both researchers and guitarists.

6. REFERENCES

[1] Jackson Loth, Pedro Sarmento, Saurjya Sarkar, and Mathieu Barthet, "Analysis of MIDI as Input Representations for Guitar Synthesis," in *DMRN+19: Digital Music Research Network One-day Workshop*, 2024.

[2] Yu-Hua Chen, Wen-Yi Hsiao, Tsu-Kuang Hsieh, Jyh-Shing Roger Jang, and Yi-Hsuan Yang, "Towards Automatic Transcription of Polyphonic Electric Guitar Music: A New Dataset and a Multi-loss Transformer Model," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

[3] Xavier Riley, Drew Edwards, and Simon Dixon, "High Resolution Guitar Transcription via Domain Adaptation," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[4] J Riley, Z Guo, D Edwards, and S Dixon, "GAPS: A Large and Diverse Classical Guitar Dataset and Benchmark Transcription Model," in *25th International Society for Music Information Retrieval (ISMIR) Conference*, 2024.

[5] Jackson Loth, Xavier Riley, Simon Dixon, and Emmanouil Benetos, "Velocity prediction in automatic guitar transcription," in *Proceedings of the 34th European Signal Processing Conference (EUSIPCO)*, Bruges, Belgium, 2026.

[6] Andrew Wiggins and Youngmoo E Kim, "Guitar Tablature Estimation with a Convolutional Neural Network.," in *Proceedings of the 20th International Society for Music Information Retrieval Conference (ISMIR)*, 2019.

[7] Frank Cwitkowitz, Toni Hirvonen, and Anssi Klapuri, "FretNet: Continuous-valued Pitch Contour Streaming for Polyphonic Guitar Tablature Transcription," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[8] Taehyeon Kim, Man-Je Kim, and Chang Wook Ahn, "Multi-task Learning-based Temporal Pattern Matching Network for Guitar Tablature Transcription," *Neural Computing and Applications*, 2025.

[9] Hegel Pedroza, Wallace Abreu, Ryan M Corey, and Iran R Roman, "Guitar-TECHS: An Electric Guitar Dataset Covering Techniques, Musical Excerpts, Chords and Scales Using a Diverse Array of Hardware," *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

[10] Riccardo Simionato and Louis Bigo, "Playability-Aware Audio-to-Tablature Guitar Transcription via Diffusion Models," in *Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026.

[11] Ralf Gunter Correa Carvalho and Paris Smaragdis, "Towards End-to-end Polyphonic Music Transcription: Transforming Music Audio Directly to a Score," in *Proceedings of IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2017.

[12] Lele Liu, Veronica Morfi, and Emmanouil Benetos, "Joint Multi-pitch Detection and Score Transcription for Polyphonic Piano Music," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.

[13] Víctor Arroyo, Jose J Valero-Mas, Jorge Calvo-Zaragoza, and Antonio Pertusa, "Neural Audio-to-Score Music Transcription for Unconstrained Polyphony Using Compact Output Representations," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

[14] Wei Zeng, Xian He, and Ye Wang, "End-to-end Real-world Polyphonic Piano Audio-to-Score Transcription with Hierarchical Decoding," *Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.

[15] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," in *International Conference on Machine Learning*, 2023.

[16] Pedro Sarmento, Adarsh Kumar, C. J. Carr, Zack Zukowski, Mathieu Barthet, and Yi-Hsuan Yang, "DadaGP: A Dataset of Tokenized GuitarPro Songs for Sequence Models," *Proceedings of the 22nd International Society for Music Information Retrieval Conference (ISMIR)*, 2021.

[17] Yongyi Zang, Yi Zhong, Frank Cwitkowitz, and Zhiyao Duan, "SynthTab: Leveraging Synthesized Data for Guitar Tablature Transcription," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

[18] J Loth, P Sarmento, S Sarkar, Z Guo, M Barthet, and M Sandler, "GOAT: A Large Dataset of Paired Guitar Audio Recordings and Tablatures," in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025.

[19] Andrew McLeod and Mark Steedman, "Evaluating Automatic Polyphonic Music Transcription," in *Proceedings of the 19th International Society for Music Information Retrieval Conference (ISMIR)*, 2018.

[20] Andrew McLeod, "Evaluating Non-aligned Musical Score Transcriptions with MV2H," *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), LBD*, 2019.

[21] Silvan David Peter and Gerhard Widmer, "Thegluenote: Learned Representations for Robust and Flexible Note Alignment," *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2024.

[22] Hendrik Schreiber, Julián Urbano, and Meinard Müller, "Music Tempo Estimation: Are We Done Yet?," *Transactions of the International Society for Music Information Retrieval*, 2020.